



Designing a financial tool in a messaging app: My work on LINE Invoice

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Skills

👤 Stakeholder Management, 📅 Product Roadmap, 👷 Product Delivery

In 2020, shortly after graduating with my bachelor's degree, I joined LINE TAIWAN as a service analyst, where I focused on improving the output quality of machine learning models. While this role improved my analytical thinking and introduced me to machine learning feature improvement, it was just the beginning.

After six months, I was given an exciting opportunity to take on a new challenge: leading the development of a brand-new product: LINE Invoice. This project marked my first end-to-end product ownership experience, and it gave me a strong foundation in user research, cross-functional collaboration, and data-driven decision-making.

In this article, I'll walk you through:

1. This is how it began
2. Why we made LINE Invoice?

Background: the **invoice ecosystem in Taiwan**

What was the problem?

3. How we knew we could solve the problem?

4. Our solution: LINE Invoice

5. After Launch - What Happened?

6. What I learned as a first-time product manager

7. Looking Ahead

1. This is how it began

Like many projects, this one started with an idea from the top.

One day, my manager shared that our CEO had pitched a new idea to the product team:

"Invoices (i.e., receipts that we get as a record after a purchase) are such a daily thing and closely tied to people's lives. Why not build an invoice tool inside LINE? It fits our vision of *Life on LINE*, right?"

It was an interesting suggestion, and yes, invoices are used every day, and LINE is one of the most commonly used apps in Taiwan. The idea made sense on the surface. But as a product manager, that's exactly when the real questions begin.

- Does it really make sense for LINE to build an invoice tool?
- Why now?
- What's our competitive advantage?
- And most importantly: how will this benefit both our users and LINE's ecosystem?

That's when the work began. It became our responsibility, not just to build the product, but to investigate whether we even should. What followed was a chain of research, market analysis, user interviews, and data deep-dives to validate the idea and help the CEO (and the entire org) make a clear, informed decision.

2. Why we made LINE Invoice?

Background: the invoice ecosystem in Taiwan

In Taiwan, every consumer transaction requires a formal invoice due to taxation laws. To encourage compliance, the government provides a popular “invoice lottery,” which has become part of Taiwanese daily life. This makes users highly motivated to store, collect, and check invoices.

To support the goal of sustainability, the government tried to digitalize invoices, and thus created a cloud (digital) invoice system, with which users can store invoices digitally via a personal mobile barcode.

What was the problem?

Even though Taiwanese users are highly motivated to collect invoices, the adoption of cloud invoices had remained low, even after more than 10 years of government promotion. As of our research phase, only about 30% of people were using the digital system.

Why? Because the user experience is not well-designed.

The existing government-provided tools were difficult to use and unintuitive:

- Most users had no idea how to apply for a mobile barcode (the core requirement for cloud invoices).
- They didn’t understand how to collect cloud invoices, or check whether they’d won a prize.
- The information was in long and complicated documents, technically available, but completely unapproachable for the average person.

So while the goal of digitization was great, the actual user journey was full of friction. For most people, it felt like a mystery, and that’s exactly why adoption stalled.

3. How we knew we could solve the problem?

Once we understood the gap in the current invoice experience in Taiwan, the next big question was,

Can we actually solve this problem better than anyone else?

This led us into a full round of market and user research. We started by analyzing the current market: how much was the volume of invoice usage (how big was the market) and what other tools were out there solving this problem?

We looked at 3-5 apps providing similar services, and tried to understand their market position, their current user base, and how much growth potential the space still had. It became clear that although these apps were functional, they weren't widely adopted. Most of them required users to download a separate app, go through a complicated onboarding process, and navigate a UI that wasn't beginner-friendly.

Next, we ran a detailed competitive UX research, walking through the full user flows of these apps, from signing up for a cloud invoice account to checking for winning invoices. Some tools were overly complex. Others didn't explain the process clearly enough. And overall, they shared the same problem: **users found the process troublesome, tedious, and not worth the effort.**

To complement our market analysis, we conducted 1-on-1 user interviews with eight participants from diverse backgrounds. Each session lasted about an hour. We asked a range of questions, from general to more specific, covering different aspects of their behavior. For example, we started with broader topics like:

- "How many invoices do you typically collect?"
- "How do you usually store or manage them?"

Then we narrowed in on cloud invoice usage, asking:

- "Do you collect or store cloud invoices?"
- "Which tools or apps do you use to manage them?"

These conversations gave us deep insights into users' awareness, habits, and pain points around both paper-based and cloud invoices.

Unsurprisingly, we uncovered that even highly motivated users had little idea how to use the digital tools available to them. Some had no clue how to even start. Many expressed their reluctance to install a new application because of security concerns or phone storage constraints. These insights gave us a clear sense of the pain points and where the current solutions were falling short.

But then came the lightbulb moment:

Over 90% of Taiwanese users already have LINE installed.

It's the go-to app for communication, and it's trusted across all age groups, including those who aren't tech-savvy. Instead of making people download yet another app, what if we could create a microservice, a simple web-based experience inside LINE? Users wouldn't have to leave the app. No storage concerns. No new logins. Just open LINE and start using it. This would make the tool not only more accessible, but also more inclusive, especially for older generations who are typically excluded from digital finance tools.

Based on all this, we moved forward with:

- A defined persona
- A clear list of pain points
- A mapped-out user journey, from first interaction to claiming their lottery prize
- A value proposition we felt confident about

The full research process itself is a story of its own, I'll be writing a separate article about our approach and methodology in more detail. But for now, what mattered most was:

We had the evidence, the insights, and a unique opportunity. It was time to turn this into something real.

4. Our solution: LINE Invoice

After diving into the research, it became clear that the pain points weren't just about functionality, they were about friction. Two main insights stood out:

- Users didn't want to download another app.
- The existing cloud invoice tools were too complicated that made it harder to start using.

So our goals were simple:

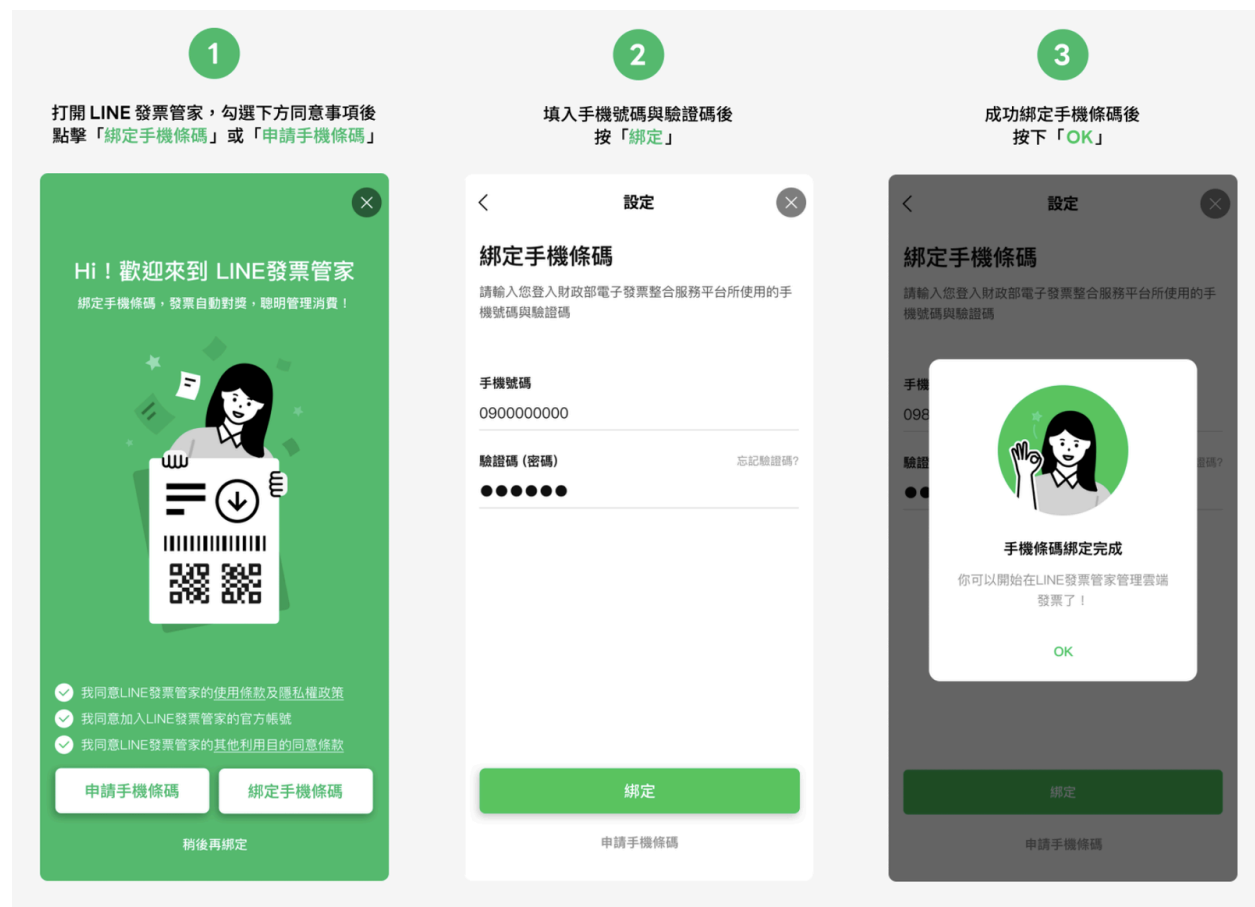
- Remove friction wherever possible, especially for first-time users.
- Strip down the experience to only what truly matters.

- And most importantly: make it all accessible inside LINE, a tool people already use and trust.

With this in mind, I started sketching out the solution. I built the first product proposal and brought it to my direct manager. Then to the CPO. Then to our CEO in Taiwan. And finally to the HQ CEO. Each round brought new feedback, challenges, and ideas, and the prototype evolved accordingly. (One day, I'll write a full article about the design thinking behind this MVP and how each decision was made. But for now, let's take a look at what the final MVP looked like.)

MVP Highlights

Seamless, Simplified Sign-Up



We knew the biggest hurdle was signing up for a cloud invoice account through the official Ministry of Finance website (財政部), a process that was confusing and lengthy.

We let users create their cloud invoice account directly within LINE Invoice. No redirection to external websites. We also created simple tutorial videos to walk users through the basics of cloud invoice, making it more welcoming and beginner-friendly.

Clear, Focused UI with Just the Essentials



Instead of cramming in every possible feature, we focused the MVP on just the core functionality.

The interface included four navigation tabs, leaving out the noises and keeping just the tools users truly needed. On the invoice list page, instead of displaying meaningless invoice numbers (which nobody reads), we showed store names and the first item purchased. This made it easier for users to identify the receipt they were looking for.

Auto-Bookkeeping (Users Loved This!)

Through user interviews, we found that many people collect invoices not only for the lottery but also for bookkeeping and spending awareness.

So we added a monthly pie chart showing a breakdown of spending categories like food, clothing, and utilities, and users loved it.

Even though it wasn't part of key invoice service functionalities, it's highly related to what the target users value. This feature became one of the most praised after the product launch. It gave users instant insight into their expenses, without needing to manually track anything.

Invoice Bingo Game

To further enhance the usage rate of cloud invoice, I incorporated a gamification element into the product, a bingo game. Most target users were interested in small rewards, even if the probability was quite low, an insight gained from the nature of invoice lottery system. We decided to add a simple game that everybody knows, Bingo, that users could receive a random prize if the invoice numbers struck a line.

The game's probability was carefully measured and designed to strike a balance between user engagement and our limited budget-as we were still a small service within the company. (If you're interested in the data-driven approach behind it, comment below and let me know! I'd be happy to share more in a separate article.)

This feature turned out to be an valuable add-on for users, and served as a special feature that further differentiated LINE Invoice from the other competitors.

Once we finalized the MVP scope and feature set, and received the green light from leadership, we moved into the next phase: UI design and development. That's a whole other chapter of the story, and one I'll cover next.

But looking back, this MVP wasn't just a stripped-down version of a bigger product. It was a deliberate, focused solution built around what real users needed, and nothing more.

5. After Launch - What Happened?

After months of research, iterations, and collaboration across teams, we launched LINE Invoice, a product that resonated far beyond our expectations. In the first year, the product attracted hundreds of thousands of users (I can't disclose the exact number), and we reached a 70% user satisfaction rate. This validated two things for us:

1. We were solving a real problem users had long struggled with.
2. We had found a unique position in the market, building a strong foundation for future product development.

6. What I learned as a first-time product manager

This project marked the beginning of my journey as a product manager, and it taught me more than I could have imagined. I learned what it really means to build a product from scratch:

To empathize with users, uncover what truly matters to them, and shape features that meet those needs clearly and meaningfully.

I also learned how to work with people from all corners of the organization:

Designers, engineers, marketers, legal and security specialists, executives. I made it my job to understand what each function values, how they think, and how to communicate with them effectively. That meant presenting information in a way that speaks their language, and collaborating in ways that helped them move forward efficiently.

I believe that being a PM is, at its core, about empathy, not just for users, but for the people you work with every day.

Thinking from others' perspectives is how we build better products. And communicating openly and clearly is how we get teams aligned around the same goal, making something truly useful.

7. Looking Ahead

This project was just the beginning.

I'm excited to share more of what I've learned in future articles, whether it's about designing the MVP, mapping user flows, running user research, or planning for further improvements on the product.

If you're interested in the journey, or curious about any specific part of the process, drop a comment or reach out, I'd love to chat. I'm always happy to connect with other builders who care deeply about products and user experience.

Thanks for reading. Until next time ✨